

Roll No.

TEC-302

B. TECH. (ECE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
DIGITAL ELECTRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Perform the following : (CO1)
(i) $(86)_9 - (77)_9$ using complement
(ii) BCD addition $792 + 565$

OR

- (b) Prove that : (CO1)
(i) $A + BC = (A + B)(A + C)$
(ii) $(A + B)(B + C)(C + A) = AB + BC + CA$

2. (a) $Y = \sum m(1, 5, 7, 13, 14, 15, 17, 18, 21, 22, 25, 29) + d(6, 9, 19, 23, 30)$.
Minimize using K-map. (CO1)

OR

- (b) $Y = \sum m(1, 3, 4, 5, 9, 10, 11) + d(6, 8)$. Minimize the expression using
Quine McCluskey method. (CO1)

P. T. O.

(2)

3. (a) Design and explain the circuit of BCD Adder. (CO2)

OR

- (b) Implement the following expression using 8 : 1 Multiplexer : (CO2)

$$Y = \sum m(1, 3, 7, 10, 11, 13, 15)$$

4. (a) Implement Full Adder using demultiplexer. (CO2)

OR

- (b) Design BCD to Excess 3 Code converter. (CO2)

5. (a) What are Flip-Flops ? Convert SR Flip-Flop into T Flip-Flop. (CO2)

OR

- (b) What is the difference between Encoder and Decoder ? Design 5 : 32 using 3 : 8 decoder. (CO2)